Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
MAKE UP EXAMINATIONS, DECEMBER 2019

ELECTRICAL CIRCUIT ANALYSIS [ELE 2153]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 20 December 2019** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. In the circuit of Fig 1A, find the value of R and hence the value of maximum power transferred to R. ***(03)***

1B. In the circuit of Fig 1B, find the current through 10Ω resistor using Norton's theorem. ***(03)***

1C. Find the response $y(t) = x(t) * h(t)$
Where $x(t) = 2u(t) - u(t-1) - u(t-2)$ and $h(t) = u(t) - u(t-3)$ ***(04)***

2A. A continuous time, aperiodic signal x(t) is shown in Fig 2A. Sketch the following:
$(i) x[-\frac{2}{3}t - 1]$ (ii) $x\left(\frac{1}{2}t\right)\partial(t-1)$ ***(03)***

2B. The output of a system is given by $y(t) = t * x(t)$, where x(t) is the input signal given by $x(t) = \cos\left(3t + \frac{\pi}{4}\right)^2 - \sin\left(5t - \frac{\pi}{8}\right) + \cos\left(\frac{2}{3}t + \frac{\pi}{3}\right)$. Check whether
(i) signal x(t) is periodic. If periodic, determine the fundamental period.
(ii) the system is stable. ***(03)***

2C. A parallel RLC circuit with R = 5Ω, L =1mH and C =10μF is excited by a current source of 10 u(t). Find the voltage across the capacitor using time domain analysis. ***(04)***

3A. In the circuit of Fig. 3A, switch is moved from A to B at t = 0, after attaining steady state at position A. Find i(t) for t > 0 using time domain analysis. ***(03)***

3B. Find the Laplace Transform of the periodic waveform shown in Fig. 3B. ***(03)***

3C. In the network of Fig. 3C, switch is changed from A to B at t = 0. Find the current through the capacitor using Laplace Transform method. ***(04)***

4A. Find the h parameters of the network shown in Fig. 4A. ***(04)***

4B. Decompose the network shown in Fig. 4B into two, 2 port networks connected in cascade and hence find the overall T parameters. ***(04)***

4C. A two port network is defined by the parameters: $Y_{21} = 6$, $Y_{22} = 8$, $h_{11} = 5$, $h_{12} = 2$. Find the T parameters. ***(02)***

5A. Write the Trigonometric Fourier Series of the waveform shown in Fig.Q5A. Also, plot the magnitude and phase spectra. ***(05)***

5B. Find the Fourier Transform of the signal shown in Fig 5B. Also, plot the magnitude spectrum. ***(03)***

5C. Find the Fourier Transform of $f(t)=e^{-2t}Cos(\pi t)u(t)$ using the properties of Fourier Transforms. ***(02)***

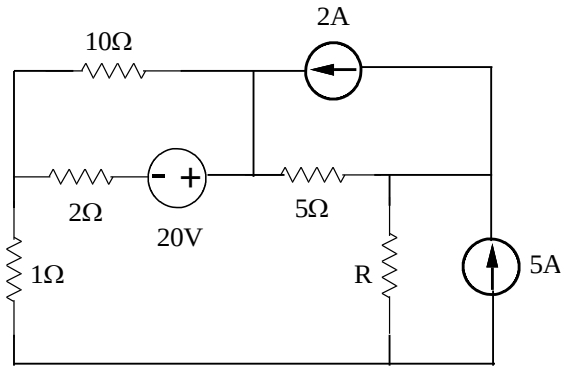


Fig 1A

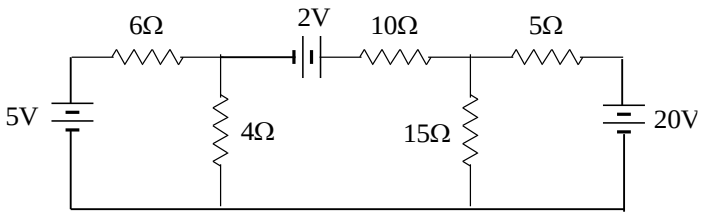


Fig 1B

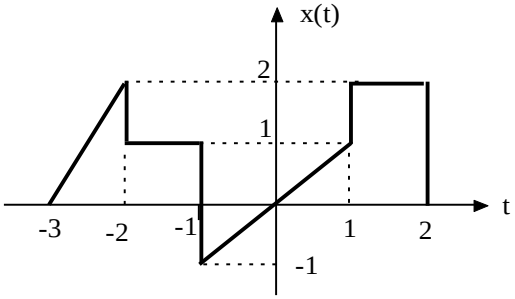


Fig 2A

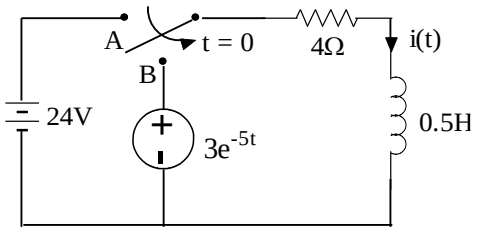


Fig. 3A

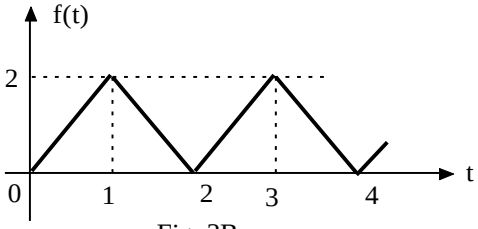


Fig. 3B

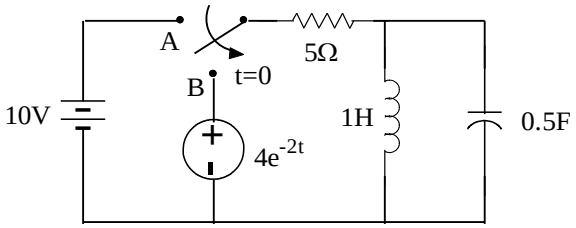


Fig. 3C

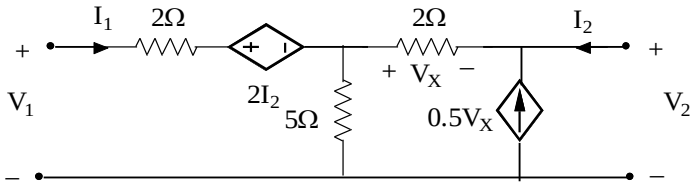


Fig. 4A

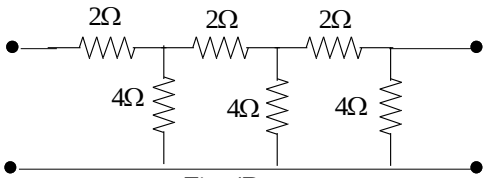


Fig. 4B

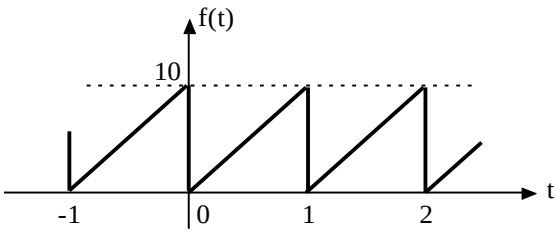


Fig 5A

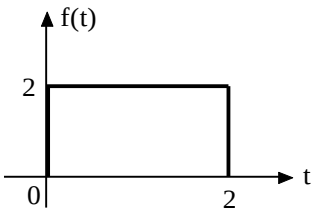


Fig 5B